Santiago Osorio Jurado

New York, NY · +1 (929) 386-4697 · santiagojurado15@hotmail.com

Portfolio · GitHub · LinkedIn · Google Scholar

RESEARCH INTERESTS

Neuromodulation for psychiatric disorders and for motor recovery after stroke and spinal cord injury, with the goal of therapies patients can use at home to restore function rather than only manage symptoms.

EDUCATION

Bachelor of Engineering (B.E.), Biomedical Engineering Aug. 2022 – May 2026
The City College of New York, CUNY | Grove School of Engineering New York, NY

- Magna Cum Laude, GPA 3.56/4.00; Dean's List; Tau Beta Pi Engineering Honor Society (top eighth of juniors, 2025).

HONORS, AWARDS & FELLOWSHIPS

NIH U-RISE Scholar (NIGMS T34), National Institutes of Health	2025 – 2026
SUPERB Summer Research Fellowship, UC Berkeley	2025
IRES Fellowship, KTH Royal Institute of Technology / CCNY	2024, 2026
Rukin Award for Academic and Professional Perseverance in BME, CCNY	2026
Wallace H. Coulter Award for Academic Service in BME, CCNY	2025, 2026
BMES Student Chapter Executive Board Award, CCNY	2025, 2026
America Needs You Fellowship	2023 – 2024

RESEARCH EXPERIENCE

CCNY Neural Engineering Group | PI: Prof. Marom Bikson New York, NY
Research Engineer Aug. 2026 – Present

- Scaling up manufacturing of the high-capacity tES electrodes used in an external 6 mA depression trial at the MUSC Brain Stimulation Lab (NCT07226011).
- Running IRB-approved tolerability sessions and improving the electrode design for the next generation.

Undergraduate Researcher Jan. 2025 – May 2026

- Engineered the multilayer Zn/AgCl redox electrodes with a hydrogel interface (≤ 600 Ω·cm) enabling safe stimulation at 6 mA, triple the conventional 2 mA dose; verified electrochemical stability via EIS, overpotential sweeps, and SEM.
- Ran IRB-approved single-blind randomized crossover sessions (n = 5; 1–6 mA), collecting VAS pain, Draize erythema, thermal imaging, and PANAS measures to produce the study's tolerability dataset.
- Developed the MRI-derived COMSOL field models (1.95–3.25 V/m cortical fields at 6 mA) and the custom PCB stimulator hardware; quantified dose-response with linear mixed-effects models in Python.

Visiting Researcher May – Aug. 2026
KTH Royal Institute of Technology | PI: Prof. Rodrigo Moreno Stockholm, Sweden

- Built a subject-specific finite-element pipeline (SimNIBS) deriving anisotropic white- and grey-matter conductivity tensors from b-tensor-encoded diffusion MRI (QTI); to our knowledge the first human tDCS field model to use MD-dMRI-derived conductivity.
- Compared three conductivity formulations (isotropic, single-shell DTI, QTI mean tensor) across four electrode montages in a 29-participant cohort (12 Parkinson's disease, 17 controls); built the workflow in Python, MATLAB, and Bash over FreeSurfer and FSL.
- Compared field predictions against MR elastography stiffness in the same cohort as an independent benchmark; first-author bioRxiv preprint under journal review.

Machine Learning Research Intern (SUPERB Fellowship) Jun. – Dec. 2025
Computational Imaging Lab, UC Berkeley | PI: Prof. Laura Waller Berkeley, CA

- Benchmarked SID (a U-Net) against SNR-Aware (a CNN-transformer hybrid) in PyTorch for denoising low-SNR darkfield microscopy; identified SID as the stronger model by PSNR, SSIM, and error maps on simulated and experimental targets.
- Trained both models from scratch on identical splits, using a 768-pair simulated dataset built from high-SNR cells (gamma correction, Poisson-Gaussian noise) and 720 experimental pairs acquired on USAF phase targets.

Medical Imaging Research Intern (IRES) May – Sep. 2024
KTH Royal Institute of Technology | PI: Prof. Mats Persson Stockholm, Sweden

- Identified 90 kVp with 40 keV virtual monoenergetic reconstruction as optimal for iodine contrast-to-noise on a Siemens NAEOTOM Alpha photon-counting CT (70–140 kVp, two dose levels, multi-energy phantoms at 0.5–20 mg/mL).
- Built an automated Python workflow comparing QIR0/QIR4 reconstruction modes and authored a physics-based parameter-selection guide for radiologists.

PUBLICATIONS & PREPRINTS

Osorio Jurado, S., Skorpil, M., Svenningsson, P., Moreno, R., & Olsson, C. (2026). Anisotropic conductivity modeling for tDCS in Parkinson's disease using multidimensional diffusion MRI. *bioRxiv.* https://doi.org/10.64898/2026.08.02.742293. Under review at *Frontiers in Human Neuroscience*.

Donnery, K., FallahRad, M., Khadka, N., Saw, M., Babaev, B., **Osorio, S.**, Belali Koochesfahani, M., Bhuiyan, R., Elwassif, J. M., & Bikson, M. (2025). High-capacity transcranial direct current stimulation (HC-tDCS). *bioRxiv.* https://doi.org/10.1101/2025.06.11.659142. Under review at *Brain Stimulation*.

SELECTED PRESENTATIONS & POSTERS

Osorio Jurado, S., Moreno, R., & Olsson, C. (2026, August). Anisotropic conductivity modeling for tDCS in Parkinson's disease using multidimensional diffusion MRI [Poster]. NYC Neuromodulation Conference, New York, NY.

Osorio Jurado, S., Moreno, R., & Olsson, C. (2026). Anisotropic conductivity modeling for tDCS in Parkinson's disease using multidimensional diffusion MRI [Poster]. Urban University Conference (UUC2026), KTH, Stockholm.

Osorio, S., Donnery, K., FallahRad, M., Belali Koochesfahani, M., Elwassif, J. M., Saw, M., Babaev, B., Bhuiyan, R., Khadka, N., & Bikson, M. (2025, November). High-capacity transcranial direct current stimulation (HC-tDCS) [Poster]. Annual Biomedical Research Conference for Minoritized Scientists (ABRCMS), San Antonio, TX.

Osorio, S. (2025, August). Deep-learning denoising for extreme short-exposure darkfield microscopy [Talk]. SUPERB Summer Research Symposium, UC Berkeley.

Osorio, S. (2024). Optimal tube-voltage selection in photon-counting CT for iodine contrast imaging [Poster]. IRES Research Symposium, KTH / CCNY.

SELECTED PROJECTS

Thermal Stimulator for Automated Rodent Pain Testing (Senior Design Capstone) Aug. 2025 – May 2026
CCNY Biomedical Engineering | Sponsor: Dr. Justin Burdge, Tactorum New York, NY

- Enabled precise, repeatable heat and cold stimuli for automated rodent pain testing, holding temperature to within ±0.5 °C and tracking a 1 °C-per-second ramp, by writing the control software that reads a temperature sensor and drives the heater through a reconfigurable four-stage cycle (cool, preheat, ramp, hold) adjustable over a serial connection while it runs.
- Restored accurate ramp tracking after the temperature consistently fell short, by tracing the fault to a control-logic reset at the constant-to-ramp handoff and retuning the feedback (PID) controller.
- Built and validated the Arduino-based power electronics that drive the heater and delivered them as a plug-in module for the sponsor's rodent-testing platform (Tactorum ARM); with a thermoelectric (Peltier) cooling stage the team added, between-test cool-down fell from 417 to 66 second.

LEADERSHIP, MENTORING & SERVICE

Chapter Officer (Secretary), Biomedical Engineering Society (BMES) 2024 – 2026
CCNY Chapter New York, NY

- Co-organized the Biodesign Challenge, a medical-device prototyping competition (56 students; $5,500 in awards) addressing real-world clinical needs.
- Mentored and placed five undergraduates, many first-generation, into faculty research positions through the ResearchConnect student-faculty matching program.

Chapter Officer (Secretary), Tau Beta Pi Engineering Honor Society 2025 – 2026
CCNY Chapter (New York Eta) New York, NY

TECHNICAL SKILLS

Programming & computing: Python (NumPy, SciPy, pandas, PyTorch, Matplotlib), MATLAB, C/C++ (Arduino), Bash; Git; LaTeX; HPC / SLURM clusters.
Neural & biomedical modeling: finite-element field modeling (COMSOL, SimNIBS); MRI processing (FreeSurfer, FSL, ANTs, SAMSEG); diffusion MRI (DTI, QTI / b-tensor encoding); anisotropic conductivity estimation; statistical modeling (mixed-effects, FDR).
Machine learning & signal processing: deep learning for image restoration and denoising in PyTorch (U-Net, CNN-transformer hybrids); synthetic data generation; PSNR/SSIM evaluation.
Electronics & devices: PCB design and stimulator electronics; electrode fabrication; electrochemical impedance spectroscopy (EIS); SEM; embedded systems (Arduino, MOSFET drivers); closed-loop PID control; 3D CAD (SolidWorks).
Experimental & translational: IRB human-subjects protocols; single-blind randomized crossover design; tolerability assays (VAS, Draize, PANAS); photon-counting CT.
Languages: Spanish (native); English (fluent, professional).